

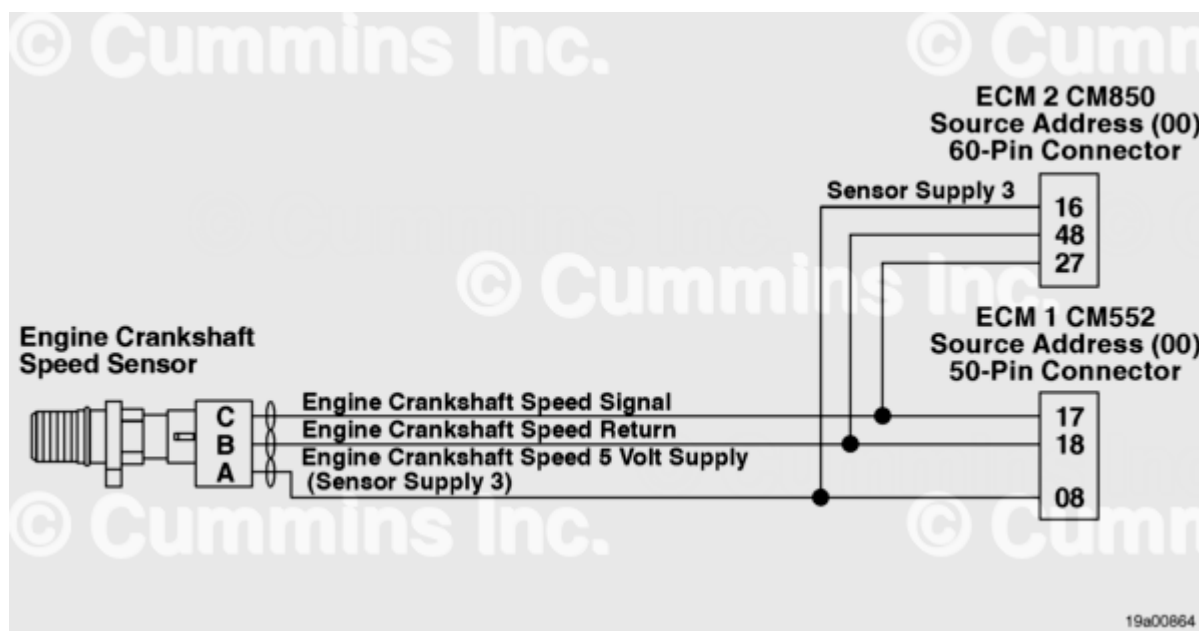
Fault Code: 689

Engine Crankshaft Speed/Position - Data Erratic, Intermittent, or Incorrect

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 689 PID(P): P190 SPN: 190 FMI: 2/2 Lamp: Amber SRT:	The ECM has detected an error in the engine speed signal.	Possible reduced performance or shutdown.



Engine Crankshaft Speed Sensor Circuit

Circuit Description

The crankshaft position and camshaft position sensors are Hall effect type sensors. The engine control module (ECM) provides a 5 volt supply to the position sensor and a return circuit. As the teeth on the crankshaft speed ring or the dimples in the back of the camshaft gear move past the

position sensor, a signal is generated on the position sensor signal circuit. The ECM interprets this signal and converts it to an engine speed. A missing tooth on the crankshaft gear is used by the ECM to determine the position of the engine.

Component Location

The engine speed sensor and the engine position sensor are located in the flywheel housing.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected a loss of the crankshaft speed/position sensor while the engine was running.

Action Taken When The Fault Code Is Active

- The generator set controller displays a warning fault immediately when the diagnostics runs and fails.
- Engine torque will be reduced if the engine is operated for an extended period of time with this fault active.

Conditions For Clearing The Fault Code

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the warning indicator immediately after the user presses reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

There are multiple ECMs for the engine models included in this manual. Each ECM has an individual source address that displays when Cummins® electronic service tool or equivalent is connected. When troubleshooting a fault code, use the source address displayed in Cummins® electronic service tool or equivalent to determine which ECM and circuit is affected.

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Damaged or malfunctioning engine speed sensor
- Damaged tone wheel

Refer to Troubleshooting Fault Code 689 . (</qs3/pubsys2/xml/en/procedures/60/60-t05-689.html>)

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